CHAPTER C

Evaluations and annual preparation

The T1 and T2 versions are available to students. Versions A and B and all corrections are reserved.
The subjects are original; they are not presented as official annals.
approaches change between versions: direct or inverse model, optimization or relative position, tree direct or overturned, projection or location.

EV1-T1 · Common DS

Training subject . Chapters 1 to 4.

Duration: 2 hours. Without calculator. Note on 20. 14 November 2026.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate 3 + 2 $\times$ 3.
   A. 12 B. 9 C. 8 D. 15

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{2}{5}$ C. $\frac{5}{6}$ D. $\frac{1}{6}$

3. Calculate $\dfrac{2^3}{2^1}$.
   A. 4 B. 8 C. 2 D. 1

4. A value of 50 increases by 20%. Final value?
   A. 10 B. 40 C. 60 D. 70

5. A value of 80 is obtained after a decrease of 20%.
   A. 60 B. 64 C. 96 D. 100

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. $\frac{9}{10}$ D. 1

7. The solution of 2 x + 3 = 0 is
   A. $\frac{3}{2}$ B. $\frac{-3}{2}$ C. 6 D. -6

8. The slope between (0; 3) and (2; 9) is
   A. 2 B. -3 C. 6 D. 3

9. Average of 3 , 5, 7 is
   A. 6 B. 15 C. 3 D. 5

10. A sector of 90 ° What fraction of the total is represented?
    A. $\frac{1}{2}$ B. $\frac{1}{4}$ C. $\frac{3}{4}$ D. $\frac{1}{3}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{2}{5}$ B. $\frac{3}{5}$ C. $\frac{5}{2}$ D. $\frac{1}{5}$

12. For $f(x) = x^2 + 3$, $f(-(2)$ is
    A. 7 B. 5 C. 12 D. -1

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Fiscal year 1 · Book fund (5 points)

A library has 120 books at the end of the reference month, rated month 0. At the end of each next month, she added 18 pounds to her fund; no books were withdrawn. u_n the number of books available at month's end n, for all natural n.

1. Determine the nature of (u_n), its common difference / ratio and expression according to n. (1 pt)

2. Calculate $u_8$then $u_0 + u_1 + \cdots + u_8$. Explain what this sum accumulates and why it does not doesn't count books all distinct. 2 pts

3. Determine the first month in which the fund strictly exceeds £300. Check the previous month. 2 pts

Exercise 2 · Area and shape choice (5 points)

A rectangle has a perimeter of 28 m. If one of its sides measures x meters, the other measure $14 - x$ meters. Its area is modelled by the function A defined on [0; 14] by $A(x) = x(14 - x)$, in metres squares. Bounds represent flattened rectangles.

1. Develop and write A in canonical form. (1 pt)

2. Determine the maximum range and the corresponding value of x without derivative. 2 pts

3. Solve $A(x) \geq 40$ on the field. 2 pts

Exercise 3 · A right angle (4 points)

The plan is equipped with an orthonormal coordinate system. $A(0; 1)$, $B(3; 2)$ and $C(-1; 4)$. lengths are expressed in the unit of the coordinate system and the areas in units of area.

1. Calculate vector coordinates $\overrightarrow{AB}$ and $\overrightarrow{AC}$. (1 pt)

2. Demonstrate that these vectors are orthogonal and specify the nature of the triangle. (1 pt)

3. Calculate length BC then the triangle ABC. 2 pts

Correction EV1-T1

MCQ: 1 B, 2 C, 3 A, 4 C, 5 D, 6 D, 7 B, 8 D, 9 D, 10 B, 11 B, 12 A. Each correct answer: 0.5 points; total 6.

Verifications of fluency. Priorities; sum of fractions; subtraction of exhibitors; multiplier coefficient 1.2; division by 0.8; product 1.25 times 0.8; affine resolution; increase vertical divided by horizontal; medium; angle divided by 360; complement to 1; substitution between parentheses.

Fiscal Year 1 · A Book Fund

1. $u_n = 120 + 18n$, arithmetical common difference 18. *Scale: 1 point. Outcome and basic justification expected; half point possible for correct approach incomplete.*
2. u_8 = 264; nine terms, sum $\frac{9(120 + 264)}{2}$ = 1728. This sum combines monthly statements, Not separate books. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*
3. $120 + 18n > 300 \iff n >$ 10. Top 11; $u_{10} = 300$, $u_{11} = 318$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 2 · Area and shape choice

1. $A = -x^2 + 14x = 49 - (x - 7)^2$. *Scale: 1 point. Expected outcome and basic justification; half point possible for correct uncompleted approach.*
2. $A \leq$ 49, equality in 7. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*
3. $49 - (x - 7)^2 \geq 40 \iff (x - 7)^2 \leq 9$, therefore [4; 10]. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 3 · A right angle

1. $\overrightarrow{AB} = (3; 1)$, $\overrightarrow{AC} = (-1; 3)$. *Scale: 1 point. Expected outcome and basic justification; half point possible for correct uncompleted approach.*
2. Product $-3 + 3 = 0$. *Scale: 1 point. Expected result and elementary justification; half point possible for correct incomplete approach.*
3. $BC = \sqrt{(-4)^2 + 2^2} = 2\sqrt{5}$; perpendicular sides of norm $\sqrt{10}$, Area 5. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of rereading. These durations are editorial estimates to be measured in class. Accept any correct method; do not sanction twice a numerical error propagated consistently.
The proposed scale does not constitute a decision to harmonise the team.

EV1-T2 · Common DS

Training subject . Chapters 1 to 4.

Duration: 2 hours. Without calculator. Note on 20. 14 November 2026.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate 4 + 2 $\times$ 3.
   A. 18 B. 9 C. 13 D. 10

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{2}{5}$ C. $\frac{5}{6}$ D. $\frac{1}{6}$

3. Calculate $\dfrac{2^4}{2^2}$.
   A. 2 B. 4 C. 1 D. 9

4. A value of 50 increases by 20%. Final value?
   A. 70 B. 40 C. 10 D. 60

5. A value of 80 is obtained after a decrease of 20%.
   A. 100 B. 96 C. 64 D. 60

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. 1 B. $\frac{21}{20}$ C. $\frac{4}{5}$ D. $\frac{9}{10}$

7. The solution of 2 x + 4 = 0 is
   A. 2 B. 8 C. -2 D. -8

8. The slope between (0; 4) and (2; 10) is
   A. 3 B. -3 C. 6 D. 2

9. Average of 4 , 6, 8 is
   A. 18 B. 6 C. 7 D. 4

10. A sector of 90 ° What fraction of the total is represented?
    A. $\frac{1}{3}$ B. $\frac{3}{4}$ C. $\frac{1}{4}$ D. $\frac{1}{2}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$

12. For $f(x) = x^2 + 4$, $f(-(2)$ is
    A. 6 B. 16 C. 8 D. 0

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Exercise 1 · Two growth processes (5 points)

Two models are compared for the production of a workshop. Day 0, each has 30 objects.
Model A provides for 10 additional objects per day; Model B provides for doubling
Daily productions of the day n are therefore $a_n = 30 + 10n$ and $b_n = 30 \times 2^n$, for $n \in \mathbb{N}$. One cumulation here means the sum of the productions of the days indicated.

1. Give the nature of each sequence and calculate the productions of the day 2. (1 pt)

2. Calculate the cumulative productions of days 0 to 3 in each model. 2 pts

3. Verify that Model B exceeds Model A on Day 1, and explain why this difference remains positive to every subsequent day by comparing daily increments. 2 pts

Exercise 2 · Imposed roots (5 points)

We're looking for a polynomial function f of degree 2, defined on $\mathbb{R}$. Its curve cuts the axis of the apses abscesses -1 and 5, and it should go through point (0; 10).
determine the function.

1. Give its factorized form. (1 pt)

2. Determine its sign and then resolve $f(x) \leq 0$. 2 pts

3. Give its canonical form and maximum. 2 pts

Exercise 3 · A controlled approximation (4 points)

The function f is defined on $\mathbb{R}$ by $f(x) = x^2 + 1$. We want to approach $f(3{,}02)$ with the aid of tangential to its curve at the point of abscisse 3. The calculations are performed without a calculator.

1. For $h \neq 0$, calculate and simplify the rate of change between 3 and $3 + h$. (1 pt)

2. Deduct from it $f'(3)$, then the tangent equation at the point of abscissa 3. (1 pt)

3. Use tangent to approach $f(3{,}02)$. Then calculate the error "exact value minus Approximation". 2 pts

EV1-T2 correction

MCQ: 1 D, 2 C, 3 B, 4 D, 5 A, 6 A, 7 C, 8 A, 9 B, 10 C, 11 D, 12 C. Each correct answer: 0.5 points; total 6.

Verifications of fluency. Priorities; sum of fractions; subtraction of exhibitors; multiplier coefficient 1.2; division by 0.8; product 1.25 times 0.8; affine resolution; increase vertical divided by horizontal; medium; angle divided by 360; complement to 1; substitution between parentheses.

Exercise 1 · Two growth processes

1. A arithmetic, B geometric; $a_2 = 50, b_2 = 120$. *Scale: 1 point. Result and justification expected elemental; half point possible for correct approach incomplete.*

2. A : $\dfrac{4(30+60)}{2} = 180$; B : $30(1+2+4+8) = 450$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

3. $b_1 = 60 > a_1$ = 40. $n \geq 1$, increase of B 30 $\times 2^n \geq 60$, greater than 10; each step increase the gap. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 2 · Imposed roots

1. $f = a(x+1)(x-5)$; $-5a$ = 10, therefore $a = -2$. *Scale: 1 point. Outcome and basic justification expected; half point possible for correct approach incomplete.*

2. Negative coefficient; f positive between roots, negative elsewhere. Solutions] $-\infty; -1] \cup [5; +\infty[$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

3. $f = -2x^2 + 8x + 10 = -2(x-2)^2$ + 18. Maximum 18 in 2. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 3 · A controlled approximation

1. Rate 6 + h. *Scale: 1 point. Expected outcome and elementary justification; half point possible for an incomplete correct approach.*

2. $f\prime(3) = 6,\ f(3) = 10$; $y = 6x - 8$. *Scale: 1 point. Expected outcome and basic justification ; half point possible for correct uncompleted approach.*

3. Approximation 10.12; exact 10 ,1204. Less approach 0 $,0004 = h^2$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of rereading. These durations are editorial estimates to be measured in class. Accept any correct method; do not sanction twice a numerical error propagated consistently.
The proposed scale does not constitute a decision to harmonise the team.

EV1-A · Common DS

Subject reserved for the professor. Chapters 1 to 4.

Duration: 2 hours. Without calculator. Note on 20. 14 November 2026.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate 5 + 2 $\times$ 3.
   A. 14 B. 11 C. 10 D. 21

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{6}$ B. $\frac{2}{5}$ C. $\frac{5}{3}$ D. $\frac{1}{6}$

3. Calculate $\dfrac{2^5}{2^3}$.
   A. 2 B. 1 C. 10 D. 4

4. A value of 50 increases by 20%. Final value?
   A. 10 B. 60 C. 70 D. 40

5. A value of 80 is obtained after a decrease of 20%.
   A. 100 B. 60 C. 64 D. 96

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{9}{10}$ B. 1 C. $\frac{21}{20}$ D. $\frac{4}{5}$

7. The solution of 2 x + 5 = 0 is
   A. 10 B. $\frac{-5}{2}$ C. -10 D. $\frac{5}{2}$

8. The slope between (0; 5) and (2; 11) is
   A. 6 B. 3 C. 2 D. -3

9. Average of 5 , 7, 9 is
   A. 8 B. 21 C. 5 D. 7

10. A sector of 90 ° What fraction of the total is represented?
    A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. $\frac{1}{4}$ D. $\frac{3}{4}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{3}{5}$ B. $\frac{2}{5}$ C. $\frac{5}{2}$ D. $\frac{1}{5}$

12. For $f(x) = x^2 + 5$, $f(-(2)$ is
    A. 9 B. 7 C. 1 D. 20

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Fiscal year 1 · Decreasing reserve (5 points)

A reserve initially contains 192 L of water. At each stage, half of the water is removed. present, without adding water. u_n the volume, in litres, remaining after n steps; row 0 corresponds to the initial state.

1. Define the sequence with its first term and common difference / ratio. (1 pt)

2. Calculate the volumes of rows 0 to 4 and their sum. 2 pts

3. Determine the first row where the volume is strictly less than 10 L; describe a loop of threshold. 2 pts

Exercise 2 · Parameter and tangent (5 points)

For a real m fixed, defined on $\mathbb{R}$ function $f_m(x) = x^2 - 2mx + 4$. The number m is a parameter. The first two questions relate to all its values; the last concerns only $m = 1$.

1. Calculate the polynomial discriminant f_m according to the m. (1 pt)

2. Determine, according to m, the number of real solutions of $f_m(x) = 0$. 2 pts

3. We're staring now. m = 1. Calculate the rate of change between 2 and 2 + h, for $h \neq 0$, then the tangent equation at the point of abscissa 2. 2 pts

Exercise 3 · Locus by the midpoint (4 points)

Two fixed points A and B the plan are distant from 8 units. I mid- segment [AB] and M a variable point of the plan. An attempt is made to describe sets of points from the product scalar $MA \cdot MB$.

1. Express $\overrightarrow{MA} \cdot \overrightarrow{MB}$ according to the MI, justifying the identity used. (1 pt)

2. Describe all points M for which this product is worth 9. (1 pt)

3. Describe successively the packages obtained when the product is worth -16, then -20. 2 pts

EV1-A correction

MCQ: 1 B, 2 A, 3 D, 4 B, 5 A, 6 B, 7 B, 8 B, 9 D, 10 C, 11 A, 12 A. Each correct answer: 0.5 point; total 6.

Verifications of fluency. Priorities; sum of fractions; subtraction of exhibitors; multiplier coefficient 1.2; division by 0.8; product 1.25 times 0.8; affine resolution; increase vertical divided by horizontal; medium; angle divided by 360; complement to 1; substitution between parentheses.

Fiscal year 1 · Decreasing reserve

1. $u_0 = 192$, $u_{n+1} = \dfrac{u_n}{2}$; step / ratio $\dfrac{1}{2}$. *Scale: 1 point. Expected outcome and basic justification; half point possible for correct uncompleted approach.*

2. $192, 96, 48, 24,$ 12, sum 372. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion : 1.*

3. $u_4 = 12$, u_5 = 6: row 5. Initialize n=0,u=192; as long as u is at least 10, divide by 2 and increase n. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 2 · Parameter and tangent

1. $\Delta = 4m^2 - 16 = 4(m^2 - 4)$. *Scale: 1 point. Expected outcome and basic justification; half point possible for correct uncompleted approach.*

2. Two roots if $m < -2$ or $m > 2$; one if $m = \pm 2$; none if $-2 < m < 2$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

3. For m=1, rate 2 + h, derivative 2 and $f(2)$ = 4 ; tangent $y = 2x$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 3 · Locus by the midpoint

1. $MI^2 -$ 16, by scalar identities. *Scale: 1 point. Outcome and basic justification expected; half point possible for correct approach incomplete.*

2. MI^2 = 25: centre circle I, radius 5. *Scale: 1 point. Outcome and basic justification expected; half point possible for correct approach incomplete.*

3. For -16, MI^2 = 0: I alone. $MI^2 = -4$: empty. *Scale: 2 points. Method or justification : 1 ; calculation and conclusion : 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of rereading. These durations are editorial estimates to be measured in class. Accept any correct method; do not sanction twice a numerical error propagated consistently.
The proposed scale does not constitute a decision to harmonise the team.

EV1-B · Common DS

Subject reserved for the professor. Chapters 1 to 4.

Duration: 2 hours. Without calculator. Note on 20. 14 November 2026.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate 6 + 2 $\times$ 3.
   A. 11 B. 12 C. 15 D. 24

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{1}{6}$ B. $\frac{5}{6}$ C. $\frac{2}{5}$ D. $\frac{5}{3}$

3. Calculate $\dfrac{2^6}{2^4}$.
   A. 2 B. 11 C. 1 D. 4

4. A value of 50 increases by 20%. Final value?
   A. 60 B. 70 C. 10 D. 40

5. A value of 80 is obtained after a decrease of 20%.
   A. 64 B. 96 C. 60 D. 100

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{21}{20}$ B. $\frac{4}{5}$ C. 1 D. $\frac{9}{10}$

7. The solution of 2 x + 6 = 0 is
   A. 3 B. 12 C. -3 D. -12

8. The slope between (0; 6) and (2; 12) is
   A. -3 B. 3 C. 6 D. 2

9. Average of 6 , 8, 10 value
   A. 8 B. 24 C. 6 D. 9

10. A sector of 90 ° What fraction of the total is represented?
    A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. $\frac{1}{4}$ D. $\frac{3}{4}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{2}{5}$ B. $\frac{3}{5}$ C. $\frac{5}{2}$ D. $\frac{1}{5}$

12. For $f(x) = x^2 + 6$, $f(-(2)$ is
    A. 8 B. 24 C. 10 D. 2

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Exercise 1 · One growth per pattern (5 points)

A family of patterns is indexed by integers $n \geq 1$. The number of points in the rank pattern n is given by $u_n = n^2 + 2n$. In particular, the first pattern comprising at least 99 points is sought; no figure is required to use the given formula.

1. Calculate u_1, u_2, u_3. (1 pt)

2. Calculate $u_{n+1} - u_n$ then establish the variation. 2 pts

3. From which rank does the pattern contain at least 99 points? Justify without infinite scanning. (2 pts)

Exercise 2 · Two dimensions and one constraint (5 points)

A rectangle has a perimeter of 26 m and an area of 40 square meters. x and y both of them These two numbers are strictly positive.
this rectangle to the other rectangles having the same perimeter.

1. Write the relationship between the two lengths x and y. (1 pt)

2. Determine the dimensions by a second degree equation. 2 pts

3. Among the rectangles of the same perimeter, determine the maximum area and compare to 40. 2 pts

Exercise 3 · Angle and then approximation (4 points)

Questions 1 and 2 concern two vectors. $\vec{u}$ and $\vec{v}$ of norms 2 and 3, such as $\vec{u} \cdot \vec{v} = 3$.
Question 3 is independent: it concerns approximation by a tangent to the square function.

1. Determine the measurement in degrees of angle not oriented between $\vec{u}$ and $\vec{v}$. (1 pt)

2. Calculate the exact value of $\|\vec{u} - \vec{v}\|$. (1 pt)

3. Independently, the tangent to the square function in 1 a for equation $y = 2x - 1$. Use for approach 1 $,03^2$, then calculate the difference between the exact value and the approximation. 2 pts

EV1-B correction

MCQ: 1 B, 2 B, 3 D, 4 A, 5 D, 6 C, 7 C, 8 B, 9 A, 10 C, 11 B, 12 C. Each correct answer: 0.5 point; total 6.

Verifications of fluency. Priorities; sum of fractions; subtraction of exhibitors; multiplier coefficient 1.2; division by 0.8; product 1.25 times 0.8; affine resolution; increase vertical divided by horizontal; medium; angle divided by 360; complement to 1; substitution between parentheses.

Exercise 1 · One growth per motive

1. $3, 8, 15$. *Scale: 1 point. Expected result and elementary justification; half point possible for correct approach incomplete.*

2. Difference 2 $n + 3 > 0$: strict growth. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

3. $n^2 + 2n - 99 = (n - 9)(n + 11) \geq 0$ with $n \geq 1 : n \geq 9$; $u_9 = 99$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 2 · Two dimensions and one constraint

1. $x + y = 13$, xy = 40, x,y positive. *Scale: 1 point. Expected outcome and basic justification; half point possible for correct uncompleted approach.*

2. Roots of $t^2 - 13t + 40 = (t - 5)(t - 8)$: 5 m and 8 m. *Scale: 2 points. Method or justification : 1 ; calculation and conclusion : 1.*

3. $A(x) = x(13 - x) = \frac{169}{4} - (x - \frac{13}{2})^2$maximum $\frac{169}{4} = 42{,}25\ \mathrm{m}^2$. Variance $\frac{9}{4}\ \mathrm{m}^2$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 3 · Angle then approximation

1. $\cos\theta = \frac{3}{6} = \frac{1}{2}$, therefore $60°$. *Scale: 1 point. Expected outcome and basic justification; half point possible for correct uncompleted approach.*

2. Square 4 + 9 − 6 = 7; norm $\sqrt{7}$. *Scale: 1 point. Expected outcome and basic justification; half point possible for correct uncompleted approach.*

3. Approximation 1.06; exact 1.0609; error 0.0009. This independent question checks the derivation Local. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of rereading. These durations are editorial estimates to be measured in class. Accept any correct method; do not sanction twice a numerical error propagated consistently.
The proposed scale does not constitute a decision to harmonise the team.

EV2-T1 · White 1

Training subject . Chapters 1 to 7.

Duration: 2 hours. Without calculator. Note on 20. 14·16 January 2027.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate 2 + 2 $\times$ 3.
   A. 8 B. 12 C. 11 D. 7

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{1}{6}$ B. $\frac{5}{3}$ C. $\frac{2}{5}$ D. $\frac{5}{6}$

3. Calculate $\dfrac{2^2}{2^0}$.
   A. 2 B. 7 C. 1 D. 4

4. A value of 50 increases by 20%. Final value?
   A. 40 B. 60 C. 70 D. 10

5. A value of 80 is obtained after a decrease of 20%.
   A. 96 B. 100 C. 64 D. 60

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{4}{5}$ B. $\frac{9}{10}$ C. 1 D. $\frac{21}{20}$

7. The solution of 2 x + 2 = 0 is
   A. -4 B. -1 C. 1 D. 4

8. The slope between (0; 2) and (2; 8) is
   A. 6 B. 2 C. 3 D. -3

9. Average of 2 , 4, 6 is
   A. 4 B. 2 C. 12 D. 5

10. A sector of 90 ° What fraction of the total is represented?
    A. $\frac{1}{3}$ B. $\frac{3}{4}$ C. $\frac{1}{4}$ D. $\frac{1}{2}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{2}{5}$ B. $\frac{5}{2}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$

12. For $f(x) = x^2 + 2$, $f(-(2)$ is
    A. 4 B. 8 C. 6 D. -2

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Fiscal year 1 · Manufacturing and control (5 points)

Two A and B workshops produce 30 per cent and 70 per cent of the objects respectively.
company. Of the A objects, 10 percent are defective; this proportion is 20 percent
at B. A random object is chosen from the total production. A and D mean
'from workshop A' and 'be defective' respectively.

1. Build a weighted tree whose first level distinguishes the two workshops. (1 pt)

2. Calculate $P(D)$, then the probability of a defective object coming from A. 2 pts

3. Events A and D Are they independent? Justify digitally. 2 pts

Exercise 2 · Minimum of a quotient (5 points)

The function f is defined on the interval]0; + ∞[by $f(x) = x + \frac{9}{x}$. We study its variations and
Some tangential to its curve. The coordinates requested are to be given in exact form.

1. Calculate and factor the derivative. (1 pt)

2. Study the variations and give the minimum. 2 pts

3. Determine the points where the tangent has a slope $\frac{3}{4}$. 2 pts

Exercise 3 · Projected and circle (4 points)

In an orthonormal coordinate system, the point is considered P(5; 2) and the right d (d) equation $x + y - 3 = 0$.
note H the orthogonal project of P on d.

1. Provide a normal non-zero vector to d. (1 pt)

2. Determine the coordinates of H and verify the conditions of belonging and orthogonality. (1 pt)

3. Give an equation of the center circle P passing through H. 2 pts

EV2-T1 correction

MCQ: 1 A, 2 D, 3 D, 4 B, 5 B, 6 C, 7 B, 8 C, 9 A, 10 C, 11 D, 12 C. Each correct answer: 0.5 point; total 6.

Verifications of fluency. Priorities; sum of fractions; subtraction of exhibitors; multiplier coefficient 1.2; division by 0.8; product 1.25 times 0.8; affine resolution; increase vertical divided by horizontal; medium; angle divided by 360; complement to 1; substitution between parentheses.

Fiscal year 1 · Manufacturing and control

1. First level: 0 ,3 and 0,7. After A: 0 ,1 and 0,9 ; after B : 0 ,2 and 0,8. Each pair sum à 1. *Scale: 1 point. Expected result and elementary justification; half point possible for correct approach incomplete.*
2. $P(D) = 0{,}03 + 0{,}14 = 0{,}17$; $P_D(A) = \dfrac{3}{17}$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*
3. No: $P_A(D) = 0{,}1 \neq 0{,}17$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 2 · Minimum of a quotient

1. $f\prime = \dfrac{(x-3)(x+3)}{x^2}$. *Scale: 1 point. Expected result and elementary justification; half point possible for correct incomplete approach.*
2. Decline up to 3 and then grow; minimum 6. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*
3. $1 - \dfrac{9}{x^2} = \dfrac{3}{4}$ gives x^2 = 36, so x=6 alone on the domain; point (6; $\dfrac{15}{2}$). *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 3 · Projected and circled

1. $(1;1)$. *Scale: 1 point. Expected result and elementary justification; half point possible for correct approach incomplete.*
2. $\lambda = \dfrac{5+2-3}{2} = 2$; H(3;0). *Scale: 1 point. Expected outcome and basic justification; half point possible for correct uncompleted approach.*
3. $PH^2 = 8$; $(x-5)^2 + (y-2)^2 = 8$. *Scale: 2 points. Method or justification: 1 ; calculation and Conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of rereading. These durations are editorial estimates to be measured in class. Accept any correct method; do not sanction twice a numerical error propagated consistently.
The proposed scale does not constitute a decision to harmonise the team.

EV2-T2 · White 1

Training subject . Chapters 1 to 7.

Duration: 2 hours. Without calculator. Note on 20. 14·16 January 2027.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate $3 + 2 \times 3$.
   A. 12 B. 8 C. 15 D. 9

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{2}{5}$ B. $\frac{1}{6}$ C. $\frac{5}{3}$ D. $\frac{5}{6}$

3. Calculate $\dfrac{2^3}{2^1}$.
   A. 4 B. 2 C. 8 D. 1

4. A value of 50 increases by 20%. Final value?
   A. 40 B. 70 C. 60 D. 10

5. A value of 80 is obtained after a decrease of 20%.
   A. 64 B. 96 C. 100 D. 60

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{9}{10}$ B. $\frac{21}{20}$ C. $\frac{4}{5}$ D. 1

7. The solution of $2x + 3 = 0$ is
   A. 6 B. -6 C. $\frac{3}{2}$ D. $\frac{-3}{2}$

8. The slope between (0; 3) and (2; 9) is
   A. -3 B. 2 C. 6 D. 3

9. Average of 3 , 5, 7 is
   A. 3 B. 5 C. 6 D. 15

10. A sector of 90 ° What fraction of the total is represented?
    A. $\frac{1}{4}$ B. $\frac{1}{3}$ C. $\frac{3}{4}$ D. $\frac{1}{2}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{2}{5}$ B. $\frac{5}{2}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$

12. For $f(x) = x^2 + 3$, $f(-(2)$ is
    A. 12 B. 7 C. -1 D. 5

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Exercise 1 · Two draws (5 points)

One urn contains four red balls and two blue balls, indistinguishable to the touch.
successively two balls at random, without putting back the first. At each draw, the balls still
The third question proposes another protocol, with remission.

1. Calculate the probability of getting two red balls. (1 pt)

2. Calculate the probability of getting one ball of each color, taking into account the two orders possible. 2 pts

3. The first ball is now handed over before the second draw. Calculate the probability of getting one ball of each color and compare it to that of the protocol without remission. 2 pts

Exercise 2 · One box (5 points)

We cut squares from the side x cm in the four corners of a square sheet of 10 cm side.
By raising the edges, one forms a box without lid. Its volume, in cubic centimetres, is
$V(x) = x(10 - 2x)^2$. This model is studied on [0; 5], including flattened bounds.

1. Calculate $V\prime$ in factorized form. (1 pt)

2. Study variations. 2 pts

3. Determine the exact maximum volume and justify that the bounds do not give the maximum. (2 pts)

Exercise 3 · Recognition and intersection (4 points)

In an orthonormal coordinate system, we consider the whole $\mathcal{C}$ of items $M(x; y)$ which verify $x^2 + y^2 + 4x - 2y - 4 = 0$. After identifying it, its intersections with a right and a tangent will be studied.

1. Show that $\mathcal{C}$ is a circle and give its center and radius. (1 pt)

2. Determine all commonalities $\mathcal{C}$ and to the right $x = -2$. (1 pt)

3. Write the equation of the tangent to the circle at the point of intersection of greater ordered. its direction. 2 pts

EV2-T2 correction

MCQ: 1 D, 2 D, 3 A, 4 C, 5 C, 6 D, 7 D, 8 D, 9 B, 10 A, 11 D, 12 B. Each correct answer: 0.5 points; total 6.

Verifications of fluency. Priorities; sum of fractions; subtraction of exhibitors; multiplier coefficient 1.2; division by 0.8; product 1.25 times 0.8; affine resolution; increase vertical divided by horizontal; medium; angle divided by 360; complement to 1; substitution between parentheses.

Exercise 1 · Two draws

1. $(\frac{4}{6})(\frac{3}{5}) = \frac{2}{5}$. *Scale: 1 point. Expected outcome and elementary justification; half point possible for an incomplete correct approach.*
2. $(\frac{4}{6})(\frac{2}{5}) + (\frac{2}{6})(\frac{4}{5}) = \frac{8}{15}$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*
3. With discount: 2($\frac{2}{3})(\frac{1}{3}) = \frac{4}{9}$. The models differ; $\frac{8}{15} > \frac{4}{9}$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 2 · A box

1. $V\prime = 4(5-x)(5-3x)$. *Scale: 1 point. Expected result and elementary justification; half point possible for correct incomplete approach.*
2. Believes until $\frac{5}{3}$ then decrease to 5. *Scale: 2 points. Method or justification: 1 ; calculation and Conclusion: 1.*
3. $V(\frac{5}{3}) = \frac{2000}{27}$; zero values at the bounds. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 3 · Recognition and intersection

1. $(x+2)^2+(y-1)^2$ = 9: centre (-2;1), radius 3. *Scale: 1 point. Outcome and basic justification expected; half point possible for correct approach incomplete.*
2. Items (-2; 4) and $(-2;-2)$. *Scale: 1 point. Expected outcome and basic justification; half point possible for correct uncompleted approach.*
3. Vertical radius at the upper point, horizontal tangent y = 4. Normal (0;1). *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of rereading. These durations are editorial estimates to be measured in class. Accept any correct method; do not sanction twice a numerical error propagated consistently.
The proposed scale does not constitute a decision to harmonise the team.

EV2-A · White 1

Subject reserved for the professor. Chapters 1 to 7.

Duration: 2 hours. Without calculator. Note on 20. 14·16 January 2027.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate 4 + 2 $\times$ 3.
   A. 13 B. 18 C. 10 D. 9

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{2}{5}$ B. $\frac{1}{6}$ C. $\frac{5}{3}$ D. $\frac{5}{6}$

3. Calculate $\dfrac{2^4}{2^2}$.
   A. 2 B. 4 C. 1 D. 9

4. A value of 50 increases by 20%. Final value?
   A. 40 B. 10 C. 60 D. 70

5. A value of 80 is obtained after a decrease of 20%.
   A. 64 B. 100 C. 96 D. 60

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{9}{10}$ B. $\frac{4}{5}$ C. 1 D. $\frac{21}{20}$

7. The solution of 2 x + 4 = 0 is
   A. 2 B. 8 C. -8 D. -2

8. The slope between (0; 4) and (2; 10) is
   A. -3 B. 3 C. 2 D. 6

9. Average of 4 , 6, 8 is
   A. 7 B. 18 C. 6 D. 4

10. A sector of 90 ° What fraction of the total is represented?
    A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{3}{4}$ D. $\frac{1}{4}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{1}{5}$ B. $\frac{3}{5}$ C. $\frac{2}{5}$ D. $\frac{5}{2}$

12. For $f(x) = x^2 + 4$, $f(-(2)$ is
    A. 16 B. 0 C. 6 D. 8

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Exercise 1 · An unknown probability (5 points)

Two events A and B of the same random experiment check $P(A) = \frac{2}{5}$, $P(B) = \frac{1}{2}$ and $P_A(B) = \frac{3}{4}$. We want to complete the probabilities of a tree whose first level distinguishes A of its supplement.

1. Calculate $P(A \cap B)$. (1 pt)

2. Determine $P_{\overline{A}}(B)$ and then complete a tree. 2 pts

3. Determine $P_B(A)$ and discuss independence. 2 pts

Fiscal year 2 · Relative position and changes (5 points)

Functions f and g are defined on [-1; 4] by $f(x) = x^3 - 3x^2 + 2$ and $g(x) = 2$. Their curves are The difference between the differences between the f the position of its curve by in relation to the g.

1. Calculate $f\prime$ and study variations. (1 pt)

2. Compare f and g curves on the domain. 2 pts

3. Determine minimum and maximum f in this area. 2 pts

Exercise 3 · One mediator (4 points)

In an orthonormal coordinate system, the points are given $A(0; 2)$, $B(6; 4)$ and $P(2; 6)$. I the midpoint of $[AB]$. We're looking for an equation of the mediator in this segment, and then we're using this line for Compare two distances.

1. Find the midpoint of AB and a normal vector to its perpendicular bisector. (1 pt)

2. Determine the mediator's equation. (1 pt)

3. Is P equidistant from A and B? Check by two methods. 2 pts

EV2-A correction

MCQ: 1 C, 2 D, 3 B, 4 C, 5 B, 6 C, 7 D, 8 B, 9 C, 10 D, 11 B, 12 D. Each correct answer: 0.5 point; total 6.

Verifications of fluency. Priorities; sum of fractions; subtraction of exhibitors; multiplier coefficient 1.2; division by 0.8; product 1.25 times 0.8; affine resolution; increase vertical divided by horizontal; medium; angle divided by 360; complement to 1; substitution between parentheses.

Exercise 1 · An unknown probability

1. $\frac{3}{10}$. *Scale: 1 point. Expected result and elementary justification; half point possible for correct approach incomplete.*
2. $P(\overline{A} \cap B) = \frac{1}{2} - \frac{3}{10} = \frac{1}{5}$; divide by $\frac{3}{5}$ gives $\frac{1}{3}$; Supplements $\frac{1}{4}$ and $\frac{2}{3}$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*
3. $P_B(A) = \frac{3}{5}$; different from $P(A) = \frac{2}{5}$, therefore not independent. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Fiscal year 2 · Relative position and changes

1. $f\prime = 3x(x-2)$; grows up to 0, decreases up to 2, then grows. *Scale: 1 point. Result and expected elementary justification; half point possible for correct incomplete approach.*
2. $f - g = x^2(x-3)$: negative for x<3 except equal in 0, zero also in 3, positive after 3. *Scale: 2 Method or justification: 1 ; calculation and conclusion: 1.*
3. $f(-1) = -2, f(0) = 2, f(2) = -2$, f(4) = 18. Minimum -2 in -1 and 2; maximum 18 in 4. *Scale : 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 3 · A mediator

1. I(3.3), normal (6;2), or (3;1). *Scale: 1 point. Expected outcome and basic justification; half point possible for correct uncompleted approach.*
2. $3x + y - 12 = 0$. *Scale: 1 point. Expected result and elementary justification; half point possible for correct incomplete approach.*
3. $3(2) + 6 - 12 = 0$ therefore yes; $PA^2 = 4 + 16 = 20$, $PB^2 = 16 + 4 = 20$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of rereading. These durations are editorial estimates to be measured in class. Accept any correct method; do not sanction twice a numerical error propagated consistently.
The proposed scale does not constitute a decision to harmonise the team.

EV2-B · White 1

Subject reserved for the professor. Chapters 1 to 7.

Duration: 2 hours. Without calculator. Note on 20. 14·16 January 2027.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate $5 + 2 \times 3$.
   A. 11 B. 10 C. 14 D. 21

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{1}{6}$ C. $\frac{5}{6}$ D. $\frac{2}{5}$

3. Calculate $\dfrac{2^5}{2^3}$.
   A. 2 B. 10 C. 1 D. 4

4. A value of 50 increases by 20%. Final value?
   A. 10 B. 60 C. 40 D. 70

5. A value of 80 is obtained after a decrease of 20%.
   A. 100 B. 64 C. 96 D. 60

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{9}{10}$ B. 1 C. $\frac{21}{20}$ D. $\frac{4}{5}$

7. The solution of $2x + 5 = 0$ is
   A. -10 B. $\frac{5}{2}$ C. $\frac{-5}{2}$ D. 10

8. The slope between (0; 5) and (2; 11) is
   A. 2 B. -3 C. 3 D. 6

9. Average of 5 , 7, 9 is
   A. 7 B. 21 C. 8 D. 5

10. A sector of 90 ° What fraction of the total is represented?
    A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. $\frac{3}{4}$ D. $\frac{1}{4}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$

12. For $f(x) = x^2 + 5$, $f(-(2)$ is
    A. 1 B. 7 C. 9 D. 20

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Exercise 1 · Independence and repetition (5 points)

Two events A and B are independent and verify $P(A) = \frac{1}{4}$ and $P(B) = \frac{2}{3}$. The first
In a separate experiment, used in questions 2 and 3, the following questions were discussed:
repeats independently three times a test whose probability of success is worth $\frac{1}{4}$.

1. Calculate $P(A \cap B)$ and $P(A \cup B)$. (1 pt)

2. Calculate the probability of exactly two successes in the three trials. 2 pts

3. Calculate that of at least one success and justify the transition to the supplementary. 2 pts

Exercise 2 · Rational and symmetry (5 points)

The function f is defined on $\mathbb{R} \setminus \{0\}$ by $f(x) = \frac{x^2 + 4}{x}$. Its domain is symmetrical in relation to
This is not an interval. It will be based on its parity to link the studies on the two
Half straight.

1. Study parity and rewrite f. (1 pt)

2. Study variations on]0; + ∞[. 2 pts

3. Deduct variations on] $-\infty$; 0[and explain why f is not increasing overall
on his field. 2 pts

Exercise 3 · Distance and parameter (4 points)

In an orthonormal coordinate system, we give the right d d) Equation $3x + 4y - 12 = 0$ and point $P(0; 0)$.
Note H the orthogonal project of P on d. We're looking for the center circle P tangential to this right.

1. Give a normal and check that (4; 0) is on d. (1 pt)

2. Calculate Project H of P. (1 pt)

3. Determine the distance from P to d and the equation of the centre circle P tangent to d. 2 pts

EV2-B correction

MCQ: 1 A, 2 C, 3 D, 4 B, 5 A, 6 B, 7 C, 8 C, 9 A, 10 D, 11 D, 12 C. Each correct answer: 0.5 points; total 6.

Verifications of fluency. Priorities; sum of fractions; subtraction of exhibitors; multiplier coefficient 1.2; division by 0.8; product 1.25 times 0.8; affine resolution; increase vertical divided by horizontal; medium; angle divided by 360; complement to 1; substitution between parentheses.

Exercise 1 · Independence and repetition

1. $\frac{1}{6}$ and $\frac{1}{4}+\frac{2}{3}-\frac{1}{6}=\frac{3}{4}$. *Scale: 1 point. Expected result and elementary justification; half point possible for correct incomplete approach.*

2. Three paths, each ($\frac{1}{4})^2(\frac{3}{4})$: total $\frac{9}{64}$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

3. Complement of three failures: 1 $-(\frac{3}{4})^3=\frac{37}{64}$. *Scale : 2 points. Method or justification : 1 ; calculation and conclusion: 1.*

Exercise 2 · Rational and symmetry

1. $f=x+\frac{4}{x}$, odd. *Scale: 1 point. Expected result and elementary justification; half point possible for correct incomplete approach.*

2. $f\prime=\frac{x^2-4}{x^2}$; decreases to 2 and then grows. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

3. Believe on $]-\infty;-2]$, decreases on $[-2;0[$. On [1; 2], f decreases: counter-example to overall growth. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 3 · Distance and Parameter

1. Normal (3;4); 12 $-12=0$. *Scale: 1 point. Expected outcome and basic justification; half point possible for correct uncompleted approach.*

2. $\lambda=-\frac{12}{25}$, therefore $H(\frac{36}{25};\frac{48}{25})$. *Scale: 1 point. Expected outcome and basic justification; half point possible for correct uncompleted approach.*

3. Distance $\frac{12}{5}$; $x^2+y^2=\frac{144}{25}$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion : 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of rereading. These durations are editorial estimates to be measured in class. Accept any correct method; do not sanction twice a numerical error propagated consistently.
The proposed scale does not constitute a decision to harmonise the team.

EV3-T1 · White 2

Training subject . Programme studied in Chapters 1 to 10.

Duration: 2 hours. Without calculator. Note on 20. 1·3 April 2027.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate 6 + 2 $\times$ 3.
   A. 12 B. 24 C. 11 D. 15

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{2}{5}$ B. $\frac{5}{3}$ C. $\frac{5}{6}$ D. $\frac{1}{6}$

3. Calculate $\dfrac{2^6}{2^4}$.
   A. 1 B. 11 C. 2 D. 4

4. A value of 50 increases by 20%. Final value?
   A. 10 B. 40 C. 70 D. 60

5. A value of 80 is obtained after a decrease of 20%.
   A. 64 B. 96 C. 100 D. 60

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{21}{20}$ B. $\frac{9}{10}$ C. 1 D. $\frac{4}{5}$

7. The solution of 2 x + 6 = 0 is
   A. 12 B. -12 C. -3 D. 3

8. The slope between (0; 6) and (2; 12) is
   A. -3 B. 3 C. 2 D. 6

9. Average of 6 , 8, 10 value
   A. 9 B. 8 C. 24 D. 6

10. A sector of 90 ° What fraction of the total is represented?
    A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{1}{4}$ D. $\frac{3}{4}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{3}{5}$ D. $\frac{1}{5}$

12. For $f(x) = x^2 + 6$, $f(-(2)$ is
    A. 10 B. 8 C. 24 D. 2

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Exercise 1 · A compound gain (5 points)

One game consists of performing two independent tests, with a probability of success of $\frac{1}{2}$ each test. Each success earns 3 monetary units; participation costs 2 units. X the net gain, i.e. total received minus interest paid.

1. Establish the distribution of probability of net gain X. (1 pt)

2. Calculate his expectation and variance. 2 pts

3. Determine the participation that would make the game fair. Explain what would become of the gap type of net gain. 2 pts

Exercise 2 · Pic d'un modèle (5 points)

A grandeur is modeled for $t \geq 0$ by function $f(t) = (t+2)e^{-t}$. Time t is expressed in a fixed unit. We look for its maximum value, then an approximation near the moment. initial.

1. Derive f. (1 pt)

2. Study the variations and give maximum. 2 pts

3. Write the tangent in 0 and approach $f(0{,}1)$. 2 pts

Exercise 3 · Trigonometric circle (4 points)

The plane is equipped with an orthonormal coordinate system. On the centre trigonometric circle O, points A and B are the respective images of the real $\frac{\pi}{6}$ and $-\frac{5\pi}{6}$.

1. Provide their contact details. (1 pt)

2. Calculate AB. (1 pt)

3. Give their mediator's equation and justify geometrically. 2 pts

EV3-T1 correction

MCQ: 1 A, 2 C, 3 D, 4 D, 5 C, 6 C, 7 C, 8 B, 9 B, 10 C, 11 C, 12 A. Each correct answer: 0.5 point; total 6.

Verifications of fluency. Priorities; sum of fractions; subtraction of exhibitors; multiplier coefficient 1.2; division by 0.8; product 1.25 times 0.8; affine resolution; increase vertical divided by horizontal; medium; angle divided by 360; complement to 1; substitution between parentheses.

Exercise 1 · A compound gain

1. Values -2,1,4; probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. *Scale: 1 point. Expected outcome and basic justification ; half point possible for correct uncompleted approach.*

2. $E(X) = 1$; $E(X^2) = 1 + \frac{1}{2} + 4 = \frac{11}{2}$; $V(X) = \frac{9}{2}$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

3. Average gross gain 3: equitable bet 3. Standard deviation unchanged $\frac{3}{\sqrt{2}}$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 2 · Pic d'un modèle

1. $f\prime = -(t+1)e^{-t}$. *Scale: 1 point. Expected result and elementary justification; half point possible for correct incomplete approach.*

2. Strictly decreasing on [0; + ∞[, maximum 2 in 0. *Scale: 2 points. Method or justification : 1 ; calculation and conclusion : 1.*

3. $y = 2 - t$; This is not an exact equality. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 3 · Trigonometric Circle

1. $A(\frac{\sqrt{3}}{2}; \frac{1}{2})$, $B(-\frac{\sqrt{3}}{2}; \frac{1}{2})$. *Scale: 1 point. Expected outcome and elementary justification; half point possible for correct incomplete approach.*

2. $AB = \sqrt{3}$. *Scale: 1 point. Expected outcome and elementary justification; half point possible for an incomplete correct approach.*

3. Midpoint $(0; \frac{1}{2})$, horizontal AB; vertical mediator $x = 0$. *Scale: 2 points. Method or justification : 1 ; calculation and conclusion : 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of rereading. These durations are editorial estimates to be measured in class. Accept any correct method; do not sanction twice a numerical error propagated consistently.
The proposed scale does not constitute a decision to harmonise the team.

EV3-T2 · White 2

Training subject . Programme studied in Chapters 1 to 10.

Duration: 2 hours. Without calculator. Note on 20. 1·3 April 2027.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate 2 + 2 $\times$ 3.
   A. 11 B. 7 C. 12 D. 8

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{1}{6}$ B. $\frac{2}{5}$ C. $\frac{5}{3}$ D. $\frac{5}{6}$

3. Calculate $\dfrac{2^2}{2^0}$.
   A. 7 B. 4 C. 2 D. 1

4. A value of 50 increases by 20%. Final value?
   A. 40 B. 70 C. 10 D. 60

5. A value of 80 is obtained after a decrease of 20%.
   A. 96 B. 60 C. 100 D. 64

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. 1 D. $\frac{9}{10}$

7. The solution of 2 x + 2 = 0 is
   A. -1 B. 4 C. -4 D. 1

8. The slope between (0; 2) and (2; 8) is
   A. 3 B. 2 C. 6 D. -3

9. Average of 2 , 4, 6 is
   A. 2 B. 4 C. 5 D. 12

10. A sector of 90 ° What fraction of the total is represented?
    A. $\frac{1}{4}$ B. $\frac{1}{2}$ C. $\frac{1}{3}$ D. $\frac{3}{4}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{5}{2}$ B. $\frac{3}{5}$ C. $\frac{1}{5}$ D. $\frac{2}{5}$

12. For $f(x) = x^2 + 2$, $f(-(2)$ is
    A. 6 B. -2 C. 4 D. 8

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Exercise 1 · Simulation and average (5 points)

Random variable X takes the values 0, 2 and 6 with the respective probabilities $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{4}$. We then simulate samples of 25 achievements independent of this variable and it is noted m the mean observed in a sample.

1. Calculating Expectation $E(X)$. (1 pt)

2. Calculate $V(X)$ and then the exact standard deviation $\sigma(X)$. 2 pts

3. Write the interval of the values of m verifier $|m - E(X)| \leq \frac{2\sigma(X)}{5}$. Explain why a simulation on a finite number of samples does not guarantee an exact percentage coverage. 2 pts

Exercise 2 · Compare two models (5 points)

We define on $\mathbb{R}$ functions $f(x) = e^{2x}$ and $g(x) = 1 + 2x$, then their difference $h = f - g$. The last question considers the values of f at full moments, from row 0.

1. Calculate the derivative of $h = f - g$. (1 pt)

2. Study h and compare curves. 2 pts

3. For $u_n = e^{2n}$, give common difference / ratio and sum of the first four terms. 2 pts

Exercise 3 · A displaced circle (4 points)

In an orthonormal coordinate system, the circle is considered $\mathcal{C}$ d) equation ($x - 1)^2 + (y + 1)^2$ = 4 and the right d (d) equation $y = -1$. Named A and B their intersection points, A being that of smaller Abscesses.

1. Give centre and radius. (1 pt)

2. Determine intersections with the right. (1 pt)

3. For M on the circle, calculate $\overrightarrow{MA} \cdot \overrightarrow{MB}$ where A and B are these intersections; justify. 2 pts

EV3-T2 correction

MCQ: 1 D, 2 D, 3 B, 4 D, 5 C, 6 C, 7 A, 8 A, 9 B, 10 A, 11 B, 12 A. Each correct answer: 0.5 points; total 6.

Verifications of fluency. Priorities; sum of fractions; subtraction of exhibitors; multiplier coefficient 1.2; division by 0.8; product 1.25 times 0.8; affine resolution; increase vertical divided by horizontal; medium; angle divided by 360; complement to 1; substitution between parentheses.

Exercise 1 · Simulation and average

1. $E(X) = 0 + \frac{1}{2} + \frac{3}{2} = 2$. *Scale: 1 point. Expected result and elementary justification; half point possible for correct incomplete approach.*
2. $E(X^2)$ = 1 + 9 = 10, variance 6, standard deviation $\sqrt{6}$. *Scale : 2 points. Method or justification : 1 ; calculation and conclusion: 1.*
3. $[2 - \frac{2\sqrt{6}}{5}; 2 + \frac{2\sqrt{6}}{5}]$. A frequency over a finite number of simulations fluctuates. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 2 · Compare two models

1. $h\prime = 2(e^{2x} - 1)$. *Scale: 1 point. Expected result and elementary justification; half point possible for correct incomplete approach.*
2. Decline to 0 and then grow; minimum 0. $f \geq g$ with equality only in 0. *Scale: 2 Method or justification: 1 ; calculation and conclusion: 1.*
3. Step / ratio e^2 ; · sum 1 + $e^2 + e^4 + e^6 = \frac{1 - e^8}{1 - e^2}$. *Scale : 2 points. Method or justification : 1 ; calculation and conclusion: 1.*

Exercise 3 · A displaced circle

1. Centre (1;-1), radius 2. *Scale: 1 point. Expected result and elementary justification; half point possible for correct incomplete approach.*
2. A(-1;-1), B(3;-1). *Scale: 1 point. Expected result and elementary justification; half point possible for correct incomplete approach.*
3. AB is a diameter, therefore produced 0, including ends; identity $MI^2 - 4 = 0$. *Scale: 2 Method or justification: 1 ; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of rereading. These durations are editorial estimates to be measured in class. Accept any correct method; do not sanction twice a numerical error propagated consistently.
The proposed scale does not constitute a decision to harmonise the team.

EV3-A · White 2

Subject reserved for the professor. Programme studied in Chapters 1 to 10.

Duration: 2 hours. Without calculator. Note on 20. 1·3 April 2027.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate $3 + 2 \times 3$.
   A. 9 B. 15 C. 8 D. 12

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{2}{5}$ B. $\frac{5}{6}$ C. $\frac{1}{6}$ D. $\frac{5}{3}$

3. Calculate $\dfrac{2^3}{2^1}$.
   A. 4 B. 1 C. 8 D. 2

4. A value of 50 increases by 20%. Final value?
   A. 10 B. 40 C. 60 D. 70

5. A value of 80 is obtained after a decrease of 20%.
   A. 96 B. 60 C. 100 D. 64

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. $\frac{9}{10}$ D. 1

7. The solution of $2x + 3 = 0$ is
   A. $\frac{3}{2}$ B. $\frac{-3}{2}$ C. -6 D. 6

8. The slope between (0; 3) and (2; 9) is
   A. 6 B. -3 C. 2 D. 3

9. Average of $3, 5, 7$ is
   A. 6 B. 15 C. 3 D. 5

10. A sector of 90°. What fraction of the total is represented?
    A. $\frac{1}{3}$ B. $\frac{1}{4}$ C. $\frac{3}{4}$ D. $\frac{1}{2}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{3}{5}$ B. $\frac{5}{2}$ C. $\frac{2}{5}$ D. $\frac{1}{5}$

12. For $f(x) = x^2 + 3$, $f(-2)$ is
    A. 12 B. 7 C. 5 D. -1

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Exercise 1 · Lottery and dispersion (5 points)

A lottery pays a gross gain G of 0, 4 or 12 monetary units with respective probabilities $\frac{1}{2}$, $\frac{3}{8}$ and $\frac{1}{8}$. To participate, we pay 3 units. $X = G - 3$ net gain.

1. Give the law of net gain X. (1 pt)

2. Calculate expectation and variance of X. 2 pts

3. We double gross earnings and bet. What become average and standard deviation of net gain? (2 pts)

Exercise 2 · Derivative and horizontal tangent (5 points)

The function f is defined on $\mathbb{R}$ by $f(x) = (2x - 3)e^x$. We study her extremum and tangent at her 0. All values are expected in exact form.

1. Calculate $f(0)$, then $f\prime(x)$ and $f\prime(0)$. (1 pt)

2. Studying variations in f on $\mathbb{R}$ and give his exact extremum. 2 pts

3. Determine the tangent equation at abscissa point 0. Specify if its point of contact is above or below the abscissa axis. 2 pts

Exercise 3 · Angle and length (4 points)

In an orthonormal coordinate system, the points are given $A(0; 0)$, $B(4; 0)$ and $C(2; 2\sqrt{3})$ We are looking for nature Triangle ABC Confronting lengths and scalar product.

1. Calculate AB and AC. (1 pt)

2. Determine the cosine and then the LAC angle in radians. (1 pt)

3. Determine BC and the nature of the triangle 2 pts

EV3-A correction

MCQ: 1 A, 2 B, 3 A, 4 C, 5 C, 6 D, 7 B, 8 D, 9 D, 10 B, 11 A, 12 B. Each correct answer: 0.5 points; total 6.

Verifications of fluency. Priorities; sum of fractions; subtraction of exhibitors; multiplier coefficient 1.2; division by 0.8; product 1.25 times 0.8; affine resolution; increase vertical divided by horizontal; medium; angle divided by 360; complement to 1; substitution between parentheses.

Exercise 1 · Lottery and dispersion

1. X is -3,1,9 with the same probability. *Scale: 1 point. Outcome and basic justification expected; half point possible for correct approach incomplete.*

2. $E(X) = -\frac{3}{2} + \frac{3}{8} + \frac{9}{8} = 0$. $E(X^2) = \frac{9}{2} + \frac{3}{8} + \frac{81}{8}$ = 15; variance 15. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

3. 2X net gain, expectation 0; standard deviation $2\sqrt{15}$. *Scale : 2 points. Method or justification : 1 ; calculation and conclusion: 1.*

Exercise 2 · Derivative and horizontal tangent

1. $f(0) = -3$; $f\prime = (2x-1)e^x$, $f\prime(0) = -1$. *Scale: 1 point. Outcome and basic justification expected; half point possible for correct approach incomplete.*

2. Decline until $\frac{1}{2}$, then grows ; minimum $-2e^{\frac{1}{2}}$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

3. $y = -x - 3$; contact (0;-3), under the horizontal axis. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 3 · Angle and length

1. AB=4, $AC = \sqrt{4+12} = 4$. *Scale: 1 point. Expected outcome and basic justification; half point possible for correct uncompleted approach.*

2. Product 8 ; cosine $\frac{8}{16} = \frac{1}{2}$, angle $\frac{\pi}{3}$. *Scale: 1 point. Outcome and basic justification expected; half point possible for correct approach incomplete.*

3. $BC = \sqrt{4+12}$ = 4: equilateral triangle. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of rereading. These durations are editorial estimates to be measured in class. Accept any correct method; do not sanction twice a numerical error propagated consistently.
The proposed scale does not constitute a decision to harmonise the team.

EV3-B · White 2

Subject reserved for the professor. Programme studied in Chapters 1 to 10.

Duration: 2 hours. Without calculator. Note on 20. 1·3 April 2027.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate $4 + 2 \times 3$.
   A. 10 B. 9 C. 18 D. 13

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{1}{6}$ B. $\frac{5}{6}$ C. $\frac{5}{3}$ D. $\frac{2}{5}$

3. Calculate $\dfrac{2^4}{2^2}$.
   A. 4 B. 1 C. 2 D. 9

4. A value of 50 increases by 20%. Final value?
   A. 70 B. 60 C. 10 D. 40

5. A value of 80 is obtained after a decrease of 20%.
   A. 96 B. 60 C. 64 D. 100

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{9}{10}$ B. 1 C. $\frac{21}{20}$ D. $\frac{4}{5}$

7. The solution of $2x + 4 = 0$ is
   A. 2 B. 8 C. -2 D. -8

8. The slope between (0; 4) and (2; 10) is
   A. 2 B. -3 C. 3 D. 6

9. Average of 4, 6, 8 is
   A. 7 B. 18 C. 6 D. 4

10. A sector of 90° What fraction of the total is represented?
    A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{3}{4}$ D. $\frac{1}{4}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{1}{5}$ B. $\frac{5}{2}$ C. $\frac{2}{5}$ D. $\frac{3}{5}$

12. For $f(x) = x^2 + 4$, $f(-2)$ is
    A. 16 B. 0 C. 6 D. 8

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Exercise 1 · Four tests and one gain (5 points)

A game consists of four independent essays; each essay succeeds with probability $\frac{1}{2}$. Gross gain G is worth 8 monetary units if the four tests pass, and 0 otherwise. A bet can be fixed to make the game fair.

1. Calculate the probability of receiving 8. (1 pt)

2. Calculate expectation and variance of G. 2 pts

3. Determine equity; explain whether expectation is a possible gain. 2 pts

Exercise 2 · A constant sign derivative (5 points)

The function is considered f defined on $\mathbb{R}$ by $f(x) = (x^2 + 1)e^x$. We want to check if a tangent horizontal necessarily corresponds to an extreme.

1. Calculate and factorize f'. (1 pt)

2. Study the variation; does the horizontal tangent in -1 give an extreme? 2 pts

3. Write the tangent in 0 and calculate $f(1)$ Exactly. 2 pts

Exercise 3 · Projection on a diagonal (4 points)

In an orthonormal coordinate system of origin O, we consider $P(2; 0)$ and the right d (d) equation $y = x$. We note H the orthogonal project of P on d. The unit vector requested at the end must be oriented to O To H.

1. Give a normal non-zero vector to the right d. (1 pt)

2. Determine the coordinates of H, then the distance PH. (1 pt)

3. Give the angle not oriented $\widehat{POH}$ in radians, then the coordinates of the vector $\dfrac{\overrightarrow{OH}}{OH}$. 2 pts

EV3-B correction

MCQ: 1 A, 2 B, 3 A, 4 B, 5 D, 6 B, 7 C, 8 C, 9 C, 10 D, 11 D, 12 D. Each correct answer: 0.5 points; total 6.

Verifications of fluency. Priorities; sum of fractions; subtraction of exhibitors; multiplier coefficient 1.2; division by 0.8; product 1.25 times 0.8; affine resolution; increase vertical divided by horizontal; medium; angle divided by 360; complement to 1; substitution between parentheses.

Exercise 1 · Four tests and one gain

1. $(\frac{1}{2})^4 = \frac{1}{16}$. *Scale: 1 point. Expected outcome and elementary justification; half point possible for an incomplete correct approach.*
2. $E(G) = \frac{1}{2}$, $E(G^2) = 4$; $V(G) = \frac{15}{4}$. *Scale: 2 points. Method or justification: 1 ; calculation and Conclusion: 1.*
3. Stake $\frac{1}{2}$. The expectation of gross gain is worth $\frac{1}{2}$, not possible on a single game. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 2 · A constant sign derivative

1. $f\prime = (x^2+2x+1)e^x = (x+1)^2e^x$. *Scale: 1 point. Expected outcome and basic justification ; half point possible for correct uncompleted approach.*
2. Positive except in -1; f increases strictly (on both sides and by connection). *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*
3. $f(0) = 1, f\prime(0) = 1 : y = x+1$; $f(1) = 2e$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 3 · Projection on a diagonal

1. Normal (1;-1). *Scale: 1 point. Expected result and elementary justification; half point possible for correct incomplete approach.*
2. H(1;1), PH= $\sqrt{2}$. *Scale: 1 point. Expected result and elementary justification; half point possible for correct incomplete approach.*
3. Angle $\frac{\pi}{4}$; vector ($\frac{\sqrt{2}}{2}; \frac{\sqrt{2}}{2}$). *Scale: 2 points. Method or justification: 1 ; calculation and conclusion : 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of rereading. These durations are editorial estimates to be measured in class. Accept any correct method; do not sanction twice a numerical error propagated consistently.
The proposed scale does not constitute a decision to harmonise the team.

EV4-T1 · EAM preparation

Training subject . The whole programme.

Duration: 2 hours. No calculator. Note of 20. May and June training; Zone EAM to be confirmed.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate 5 + 2 $\times$ 3.
   A. 14 B. 11 C. 21 D. 10

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{5}{6}$ C. $\frac{1}{6}$ D. $\frac{2}{5}$

3. Calculate $\dfrac{2^5}{2^3}$.
   A. 1 B. 4 C. 10 D. 2

4. A value of 50 increases by 20%. Final value?
   A. 40 B. 10 C. 60 D. 70

5. A value of 80 is obtained after a decrease of 20%.
   A. 60 B. 96 C. 100 D. 64

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{4}{5}$ B. 1 C. $\frac{9}{10}$ D. $\frac{21}{20}$

7. The solution of 2 x + 5 = 0 is
   A. $\frac{5}{2}$ B. 10 C. $\frac{-5}{2}$ D. -10

8. The slope between (0; 5) and (2; 11) is
   A. 3 B. -3 C. 6 D. 2

9. Average of 5 , 7, 9 is
   A. 7 B. 21 C. 5 D. 8

10. A sector of 90 ° What fraction of the total is represented?
    A. $\frac{1}{4}$ B. $\frac{3}{4}$ C. $\frac{1}{3}$ D. $\frac{1}{2}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{3}{5}$ B. $\frac{1}{5}$ C. $\frac{2}{5}$ D. $\frac{5}{2}$

12. For $f(x) = x^2 + 5$, $f(-(2)$ is
    A. 20 B. 7 C. 1 D. 9

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Exercise 1 · Select a law (5 points)

We compare two games. The first gives a win. $X = 2$ with certainty. The second gives a gain $Y = -1$ with probability $\frac{2}{3}$ and one gain $Y = 8$ with probability $\frac{1}{3}$. Gains are expressed in the the same monetary unit; a negative value represents a loss.

1. Calculate $E(X)$ and $E(Y)$. (1 pt)

2. Calculate $V(Y)$ and compare the dispersions of the two gains. 2 pts

3. We play both games and add the winnings. $E(X+Y)$ by quoting the property used. 2 pts

Exercise 2 · Discreet Threshold and Continuous Model (5 points)

A quantity is modelled by $Q(t) = 64e^{-t}$ for $t \geq 0$. It is observed at full moments $n \in \mathbb{N}$ and we set $u_n = Q(n)$ For comparisons without calculators, the framework is given $e^2 < 8 < e^3$.

1. Study the direction of variation. (1 pt)

2. Determine the first integer n as $Q(n) < 8$ with the help of the guidance provided. 2 pts

3. Give the nature of the $u_n = Q(n)$ and expression of $u_0 + \cdots + u_3$. 2 pts

Exercise 3 · Equation and circle (4 points)

In an orthonormal coordinate system, we consider the whole $\mathcal{C}$ defined by $x^2 + y^2 - 2x - 4y + 1 = 0$. determine its characteristics and then identify a particular tangent.

1. Show that $\mathcal{C}$ is a circle and specifies its center and radius. (1 pt)

2. Determine its intersection points with the equation line $x = 1$. (1 pt)

3. Determine the equation of the tangent to the circle at the point of greater order and justify. (2 pts)

EV4-T1 correction

MCQ: 1 B, 2 B, 3 B, 4 C, 5 C, 6 B, 7 C, 8 A, 9 A, 10 A, 11 A, 12 D. Each correct answer: 0.5 point; total 6.

Verifications of fluency. Priorities; sum of fractions; subtraction of exhibitors; multiplier coefficient 1.2; division by 0.8; product 1.25 times 0.8; affine resolution; increase vertical divided by horizontal; medium; angle divided by 360; complement to 1; substitution between parentheses.

Exercise 1 · Choosing a law

1. $E(X) = 2$; $E(Y) = -\frac{2}{3} + \frac{8}{3} = 2$. *Scale: 1 point. Expected outcome and basic justification ; half point possible for correct uncompleted approach.*
2. $E(Y^2) = \frac{2}{3} + \frac{64}{3}$ = 22, variance 18, compared to 0 for X. *Scale: 2 points. Method or justification : 1 ; calculation and conclusion : 1.*
3. Linearity: expectation 4. X is also constant. *Scale : 2 points. Method or justification : 1 ; calculation and conclusion: 1.*

Exercise 2 · Discreet Threshold and Continuous Model

1. $Q\prime = -64e^{-t} < 0$. *Scale: 1 point. Expected result and elementary justification; half point possible for correct incomplete approach.*
2. $Q(2) > 8$, $Q(3) < 8$; decrease: first row 3. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*
3. Geometric, first 64, common ratio e^{-1} ; sum $\frac{64(1-e^{-4})}{1-e^{-1}}$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 3 · Equation and circle

1. $(x-1)^2+(y-2)^2$ = 4; centre (1;2), radius 2. *Scale: 1 point. Outcome and basic justification expected; half point possible for correct approach incomplete.*
2. (1;0) and (1;4). *Scale: 1 point. Expected outcome and elementary justification; half point possible for an incomplete correct approach.*
3. Vertical radius at point (1;4), horizontal tangent $y = 4$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of rereading. These durations are editorial estimates to be measured in class. Accept any correct method; do not sanction twice a numerical error propagated consistently.
The proposed scale does not constitute a decision to harmonise the team.

EV4-T2 · EAM Preparation

Training subject . The whole programme.

Duration: 2 hours. No calculator. Note of 20. May and June training; Zone EAM to be confirmed.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate 6 + 2 $\times$ 3.
   A. 24 B. 15 C. 11 D. 12

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{6}$ B. $\frac{2}{5}$ C. $\frac{5}{3}$ D. $\frac{1}{6}$

3. Calculate $\dfrac{2^6}{2^4}$.
   A. 2 B. 4 C. 1 D. 11

4. A value of 50 increases by 20%. Final value?
   A. 60 B. 10 C. 70 D. 40

5. A value of 80 is obtained after a decrease of 20%.
   A. 64 B. 100 C. 96 D. 60

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. 1 D. $\frac{9}{10}$

7. The solution of 2 x + 6 = 0 is
   A. -3 B. 12 C. 3 D. -12

8. The slope between (0; 6) and (2; 12) is
   A. 2 B. -3 C. 3 D. 6

9. Average of 6 , 8, 10 value
   A. 6 B. 24 C. 8 D. 9

10. A sector of 90 ° What fraction of the total is represented?
    A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. $\frac{3}{4}$ D. $\frac{1}{4}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$

12. For $f(x) = x^2 + 6$, $f(-(2)$ is
    A. 10 B. 8 C. 24 D. 2

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Exercise 1 · One overturned tree (5 points)

Two events A and B check $P(A) = \frac{3}{5}$, $P_A(B) = \frac{1}{3}$ and $P_{\overline{A}}(B) = \frac{3}{4}$. We're trying to build a tree that starts by distinguishing B and $\overline{B}$, for A and $\overline{A}$.

1. Calculate $P(A \cap B)$ and $P(B)$. (1 pt)

2. Calculate $P_B(A)$ and $P_{\overline{B}}(A)$. 2 pts

3. Build and complete the weighted tree whose first level distinguishes B and $\overline{B}$, then the second A and $\overline{A}$. 2 pts

Exercise 2 · Two optimisation methods (5 points)

The function f is defined on the interval [1; 6] by $f(x) = x(8 - x)$ We're looking for his extrema on this field, then the values of x for which $f(x)$ reached at least 15.

1. Give a canonical form and maximum. (1 pt)

2. Retrieve maximum by bypass. 2 pts

3. Determine the minimum in the field and resolve $f(x) \geq 15$. 2 pts

Exercise 3 · Arguments and symmetry (4 points)

In an orthonormal coordinate system of origin O, points A and B of the trigonometric circle are the images of $\frac{7\pi}{6}$ and $11\frac{\pi}{6}$. Unoriented angle $\widehat{AOB}$ will be given in radians, between 0 and π.

1. Give their exact coordinates. (1 pt)

2. Calculate $\overrightarrow{OA} \cdot \overrightarrow{OB}$. (1 pt)

3. Determine the angle not oriented AOB then AB. 2 pts

EV4-T2 correction

MCQ: 1 D, 2 A, 3 B, 4 A, 5 B, 6 C, 7 A, 8 C, 9 C, 10 D, 11 D, 12 A. Each correct answer: 0.5 points; total 6.

Verifications of fluency. Priorities; sum of fractions; subtraction of exhibitors; multiplier coefficient 1.2; division by 0.8; product 1.25 times 0.8; affine resolution; increase vertical divided by horizontal; medium; angle divided by 360; complement to 1; substitution between parentheses.

Exercise 1 · An overturned tree

1. $P(A \cap B) = \frac{1}{5}$; $P(B) = \frac{1}{5} + \frac{3}{10} = \frac{1}{2}$. *Scale: 1 point. Outcome and basic justification expected; half point possible for correct approach incomplete.*
2. $P_B(A) = \frac{2}{5}$; $P(A \cap \overline{B}) = \frac{2}{5}$ so $P_{\overline{B}}(A) = \frac{4}{5}$. *Scale : 2 points. Method or justification : 1 ; calculation and conclusion: 1.*
3. First level $\frac{1}{2}, \frac{1}{2}$; after B: $\frac{2}{5}$ and $\frac{3}{5}$; after addition to B: $\frac{4}{5}$ and $\frac{1}{5}$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 2 · Two optimisation methods

1. $f = 16 - (x-4)^2$; maximum 16 in 4. *Scale: 1 point. Outcome and basic justification expected; half point possible for correct approach incomplete.*
2. $f\prime = 8 - 2x$, positive before 4 and negative after. *Scale : 2 points. Method or justification : 1 ; calculation and conclusion: 1.*
3. Bornes: f(1)=7, f(6)=12; minimum 7. $x-4)^2 \leq 1 : [3;5]$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 3 · Arguments and symmetry

1. $A(-\frac{\sqrt{3}}{2}; -\frac{1}{2})$, $B(\frac{\sqrt{3}}{2}; -\frac{1}{2})$. *Scale: 1 point. Expected outcome and basic justification; half point possible for correct uncompleted approach.*
2. $-\frac{3}{4} + \frac{1}{4} = -\frac{1}{2}$. *Scale: 1 point. Expected result and elementary justification; half point possible for correct incomplete approach.*
3. Angle $\frac{2\pi}{3}$; $AB = \sqrt{3}$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of rereading. These durations are editorial estimates to be measured in class. Accept any correct method; do not sanction twice a numerical error propagated consistently.
The proposed scale does not constitute a decision to harmonise the team.

EV4-A · EAM Preparation

Subject reserved for the professor. The whole programme.

Duration: 2 hours. No calculator. Note of 20. May and June training; Zone EAM to be confirmed.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate $2 + 2 \times 3$.
   A. 11 B. 7 C. 8 D. 12

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{1}{6}$ C. $\frac{5}{6}$ D. $\frac{2}{5}$

3. Calculate $\dfrac{2^2}{2^0}$.
   A. 2 B. 1 C. 7 D. 4

4. A value of 50 increases by 20%. Final value?
   A. 70 B. 40 C. 60 D. 10

5. A value of 80 is obtained after a decrease of 20%.
   A. 100 B. 60 C. 96 D. 64

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. 1 B. $\frac{4}{5}$ C. $\frac{21}{20}$ D. $\frac{9}{10}$

7. The solution of $2x + 2 = 0$ is
   A. 1 B. 4 C. -1 D. -4

8. The slope between (0; 2) and (2; 8) is
   A. -3 B. 3 C. 6 D. 2

9. Average of 2, 4, 6 is
   A. 4 B. 5 C. 2 D. 12

10. A sector of 90° What fraction of the total is represented?
    A. $\frac{1}{4}$ B. $\frac{1}{3}$ C. $\frac{3}{4}$ D. $\frac{1}{2}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{5}{2}$ B. $\frac{3}{5}$ C. $\frac{2}{5}$ D. $\frac{1}{5}$

12. For $f(x) = x^2 + 2$, $f(-2)$ is
    A. 6 B. -2 C. 4 D. 8

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Exercise 1 · Two related variables (5 points)

Random variable X takes values 1, 2 and 3, each with probability $\frac{1}{3}$. We define on the same experience variable $Y = 4 - X$. We want to check whether a valid rule for expectation remains true for variance.

1. Calculate $E(X)$ and $E(Y)$. (1 pt)

2. Calculate $V(X)$, $V(Y)$ and $V(X+Y)$. 2 pts

3. Use these results to refute the statement: "The variance of a sum is always equal to the sum of the variances." 2 pts

Exercise 2 · Compare curve and straight (5 points)

Functions $f(x) = e^{-x}$ and $g(x) = 1 - x$ are defined on $\mathbb{R}$. We're laying $h = f - g$. The comparison of their curves will then allow to control the direction of an approximation.

1. Derive their difference. (1 pt)

2. Prove $f(x) \geq g(x)$ for all real x. 2 pts

3. Write the affine approximation of $e^{-0{,}02}$ and whether it is by default or by excess. 2 pts

Exercise 3 · Normal vector and radius (4 points)

In an orthonormal coordinate system, the point is considered $I(1; 1)$ and the right d d) Equation $3x + 4y - 17 = 0$. Note H the orthogonal project of I on d. We're looking for the center circle I tangential to d.

1. Calculate the distance from I to d. (1 pt)

2. Calculate Project H. (1 pt)

3. Give the equation of the centre circle I tangential to d and verify that H belongs to the circle. (2 pts)

EV4-A correction

MCQ: 1 C, 2 C, 3 D, 4 C, 5 A, 6 A, 7 C, 8 B, 9 A, 10 A, 11 B, 12 A. Each correct answer: 0.5 points; total 6.

Verifications of fluency. Priorities; sum of fractions; subtraction of exhibitors; multiplier coefficient 1.2; division by 0.8; product 1.25 times 0.8; affine resolution; increase vertical divided by horizontal; medium; angle divided by 360; complement to 1; substitution between parentheses.

Exercise 1 · Two related variables

1. The two expectations are worth two. *Scale: 1 point. Expected outcome and basic justification; half point possible for correct uncompleted approach.*

2. $E(X^2) = \frac{14}{3}$, therefore $V(X) = \frac{2}{3}$; same variance for Y. X+Y=4, variance 0. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

3. $0 \neq \frac{2}{3} + \frac{2}{3}$. The counter example is on the same universe; Y depends on X. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 2 · Compare curve and straight

1. $h\prime = 1 - e^{-x}$. *Scale: 1 point. Expected outcome and elementary justification; half point possible for an incomplete correct approach.*

2. Negative if x<0, positive if x>0; minimum h(0)=0. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

3. Approximation 0.98; default due to inequality. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 3 · Normal Vector and Radius

1. $\frac{|3 + 4 - 17|}{5} = 2$. *Scale: 1 point. Expected result and elementary justification; half point possible for correct incomplete approach.*

2. $\lambda = -\frac{10}{25} = -\frac{2}{5}$; H$(\frac{11}{5}; \frac{13}{5})$. *Scale: 1 point. Expected outcome and basic justification; half point possible for correct uncompleted approach.*

3. $(x-1)^2 + (y-1)^2$ = 4; for H, $\frac{6}{5})^2 + (\frac{8}{5})^2 = 4$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of rereading. These durations are editorial estimates to be measured in class. Accept any correct method; do not sanction twice a numerical error propagated consistently.
The proposed scale does not constitute a decision to harmonise the team.

EV4-B · EAM Preparation

Subject reserved for the professor. The whole programme.

Duration: 2 hours. No calculator. Note of 20. May and June training; Zone EAM to be confirmed.

Part 1 · Fluency (6 points)

Twelve questions, 0.5 points each. Only one correct answer; no justification requested.
incorrect or missing answer returns 0, without withdrawal of points.

1. Calculate $3 + 2 \times 3$.
   A. 9 B. 12 C. 15 D. 8

2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{5}{6}$ C. $\frac{2}{5}$ D. $\frac{1}{6}$

3. Calculate $\dfrac{2^3}{2^1}$.
   A. 8 B. 2 C. 4 D. 1

4. A value of 50 increases by 20%. Final value?
   A. 70 B. 40 C. 60 D. 10

5. A value of 80 is obtained after a decrease of 20%.
   A. 60 B. 64 C. 96 D. 100

6. Increase by 25% and then decrease by 20%: overall coefficient?
   A. $\frac{21}{20}$ B. 1 C. $\frac{4}{5}$ D. $\frac{9}{10}$

7. The solution of $2x + 3 = 0$ is
   A. $\frac{3}{2}$ B. 6 C. $\frac{-3}{2}$ D. -6

8. The slope between (0; 3) and (2; 9) is
   A. 6 B. -3 C. 3 D. 2

9. Average of 3, 5, 7 is
   A. 3 B. 15 C. 6 D. 5

10. A sector of 90° What fraction of the total is represented?
    A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. $\frac{1}{4}$ D. $\frac{3}{4}$

11. Probability $\dfrac{2}{5}$. Its complement is valid
    A. $\frac{3}{5}$ B. $\frac{1}{5}$ C. $\frac{2}{5}$ D. $\frac{5}{2}$

12. For $f(x) = x^2 + 3$, $f(-(2)$ is
    A. 12 B. 5 C. -1 D. 7

Part 2 · Independent exercises (14 points)

All steps must be justified; the exact results are expected unless otherwise indicated.

Exercise 1 · Average rate (5 points)

One service offers several rates. 60 per cent of customers choose Formula A; among them, half pays 0 units and the other half pays 4 units. All other customers pay 6 units. selects a customer at random and notes X the price he pays. The last question looks at a reduction uniform, including for the initially zero tariff.

1. Establish the distribution of probability of X. (1 pt)

2. Calculate his expectation and variance. 2 pts

3. All prices decrease by 1 unit. Calculate the new average and the new standard deviation. the negative price obtained for some customers. 2 pts

Exercise 2 · exponential maximum (5 points)

The function f is defined on [0; 4] by $f(t) = t^2e^{-t}$. We're looking for his maximum over this interval. the last comparison without calculator, we can use the given inequality $e^2 > 4$.

1. Calculate $f\prime$ in factorized form. (1 pt)

2. Determine the variations and maximum. 2 pts

3. Compare exactly $f(4)$ at most, knowing $e^2 > 4$. 2 pts

Exercise 3 · Determine a position (4 points)

In an orthonormal coordinate system, the points $A(0; 0)$ and $B(4; 0)$ are fixed. M moves on the right of equation $y = 2$; its coordinates are therefore $M(x; (2), \text{where } x$ travel $\mathbb{R}$.

1. Calculate $\overrightarrow{MA} \cdot \overrightarrow{MB}$. (1 pt)

2. Determine the x for which both vectors are orthogonal. (1 pt)

3. Interpret the result using the AB diameter circle. 2 pts

EV4-B correction

MCQ: 1 A, 2 B, 3 C, 4 C, 5 D, 6 B, 7 C, 8 C, 9 D, 10 C, 11 A, 12 D. Each correct answer: 0.5 points; total 6.

Verifications of fluency. Priorities; sum of fractions; subtraction of exhibitors; multiplier coefficient 1.2; division by 0.8; product 1.25 times 0.8; affine resolution; increase vertical divided by horizontal; medium; angle divided by 360; complement to 1; substitution between parentheses.

Exercise 1 · Average rate

1. Values 0.4.6; probabilities 0.3; 0.3; 0.4. *Scale: 1 point. Outcome and basic justification expected; half point possible for correct approach incomplete.*

2. $E(X) = 3{,}6 = \frac{18}{5}$; $E(X^2) = 19{,}2 = \frac{96}{5}$; variance $\frac{156}{25}$. *Scale: 2 points. Method or justification : 1 ; calculation and conclusion : 1.*

3. Average $\frac{13}{5}$; standard deviation $\frac{2\sqrt{39}}{5}$, unchanged. The price -1 would be a refund: check if This policy is really planned. *Scale: 2 points. Method or justification: 1 ; calculation and Conclusion: 1.*

Exercise 2 · exponential maximum

1. $f\prime = t(2-t)e^{-t}$. *Scale: 1 point. Expected result and elementary justification; half point possible for correct incomplete approach.*

2. Increase from 0 to 2 and then decrease; maximum $4e^{-2}$. *Scale : 2 points. Method or justification : 1 ; calculation and conclusion: 1.*

3. $f(4) = 16e^{-4}$; ratio to maximum 4 $e^{-2} < 1$ since $e^2 > 4$. *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Exercise 3 · Determine a position

1. $(-x;-2) \cdot (4-x;-2) = x^2 - 4x + 4 = (x-2)^2$. *Scale: 1 point. Result and justification expected elemental; half point possible for correct approach incomplete.*

2. Zero product only in x=2. *Scale: 1 point. Expected outcome and basic justification; half point possible for correct uncompleted approach.*

3. Centre circle (2.0), radius 2; the right y=2 is tangent to it in (2;2). *Scale: 2 points. Method or justification: 1 ; calculation and conclusion: 1.*

Calibration. Target time: 20·25 min for fluency, 80·85 min for exercises, 10·15 min of rereading. These durations are editorial estimates to be measured in class. Accept any correct method; do not sanction twice a numerical error propagated consistently.
The proposed scale does not constitute a decision to harmonise the team.